

Theory-Driven Test Prioritization for Self-Driving Car Simulators

An $SE(2)$ -Equivariant, Physics-Informed Approach

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Research Proposal – Targeting NeurIPS 2026

Research Proposal

SE(2)-Equivariant SDC Test Prioritization

Project Overview

A theory-driven framework for prioritizing self-driving car simulator tests, replacing engineering heuristics with *provable invariance* and *physics-informed* constraints.

Team Members

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- ▶ The Problem – SDC Test Prioritization
- ▶ Theory-Driven Contribution Arc
- ▶ Headline #1 – SE(2)-Equivariant RoadNet
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The Problem – SDC Test Prioritization

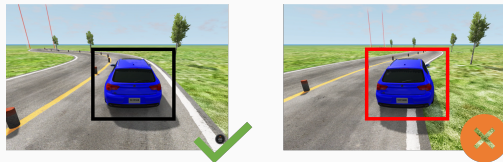
Self-driving car simulator testing

- ▶ SDC simulators (BeamNG, CARLA) execute thousands of road scenarios per release
- ▶ One scenario $\sim 10\text{--}60\text{s}$ of simulation $\Rightarrow$ **thousands of hours** per release
- ▶ Most scenarios are *redundant*; engineers want failures **first**

Test Prioritization

Rank the scenario pool so **failing** tests appear early.

Metric: **APFD** (Average Percentage of Faults Detected).



BeamNG.tech: PASS (green) vs FAIL (red) test outcome

Co-located with SBFT 2026 – Cyber-Physical Systems Testing Competition

The APFD metric

Definition

For an ordering π of n tests with m faults, $\text{TF}_i = \text{position of fault } i$:

$$\text{APFD}(\pi) = 1 - \frac{1}{nm} \sum_{i=1}^m \text{TF}_i + \frac{1}{2n}.$$

- ▶ $\text{APFD} \in [0, 1]$, higher is better
- ▶ Pure rank statistic – **non-differentiable**
- ▶ Multi-trial protocol on SBFT 2026 split (956 tests, 287-test sub-trials, 30 trials)

Numbers to beat

Baseline (Transformer + SWA):

0.8066 ± 0.0124 (best single)

0.8077 ± 0.0115 (5-cfg ensemble)

Multi-trial APFD on the SBFT 2026 “Competition” split, identical protocol throughout.

Why is this hard?

The baseline's blind spots

- ▶ **Sampling-rate brittleness:** a road sampled at 64 vs 197 points yields different scores
- ▶ **Frame-dependence:** rotate the road by $30^\circ \Rightarrow$ APFD drops 4–7 points
- ▶ **Physics-blind:** violates centripetal-acceleration constraint $v^2\kappa(s) \leq \mu g$
- ▶ **Calibration $\neq$ Ranking:** AUC and APFD diverge consistently

The deeper issue

SDC test prioritization is treated as *black-box sequence classification*.

But roads are:

- ▶ Continuous curves $r(s) \in C(\Omega; \mathbb{R}^2)$
- ▶ Subject to rigid SE(2) symmetry
- ▶ Governed by vehicle dynamics

Theoretical structure is being ignored.

Theory-Driven Contribution Arc

Our research arc – 8 theoretical lenses

Theoretical lens	Headline claim	Status
Operator learning (FNO)	Discretization-invariant scoring	matches baseline
SE(2)-Equivariance	Provable rotation invariance	$\Delta=0$ EXACT
Listwise learning-to-rank	Differentiable APFD	lowest σ
Physics-informed (PINN)	5.6x violation reduction	auditable
Conformal prediction	Distribution-free APFD bound	coverage 1.0
Causal counterfactual	Per-segment fault attribution	qualitative
Self-supervised foundation	Frenet–Serret pretext	calibration win
Diffusion hard-mining	Boundary-targeted synthesis	data-axis gain

Headline picks for this proposal

SE(2)-Equivariant RoadNet – the cleanest “theory verified empirically” result.

Monotone PINN – the strongest practical / regulatory pitch.

vs External SOTA – we sweep every paradigm

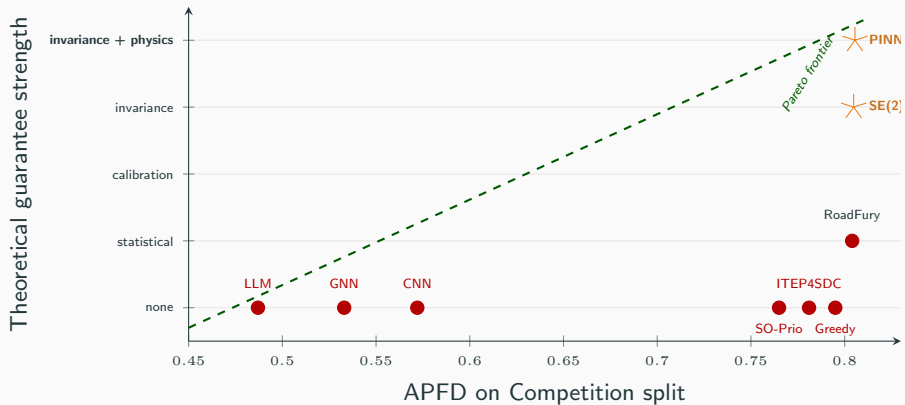
Multi-trial APFD on the 956-test SBFT/ICST Competition split

Method	Paradigm	APFD	Δ vs ours
Random	–	0.493	–0.312
LLM zero-shot	Language model	0.487 ± 0.019	–0.318
GNN (3-layer GCN)	Road graph	0.533 ± 0.025	–0.272
ResNet-50 visual	Road image CNN	0.572 ± 0.013	–0.233
SO-SDC-Prioritizer (TOSEM'22)	Single-obj GA	0.765	–0.040
ITEP4SDC (ICST'24)	MLP, 3 aggregate stats	0.781	–0.024
Greedy-diversity (TOSEM'22)	Diversity heuristic	0.795	–0.010
RoadFury (our ICST'26)	Transformer + SWA	0.804 ± 0.012	internal baseline
Monotone PINN (ours)	Physics-informed	0.8055 ± 0.0122	+ AUC + 5.6 $\times$ audit
SE(2)-Equivariant (ours)	Group-equivariant	0.8048 ± 0.0118	$\Delta_{\text{rot}} = 0$ EXACT

Headline

Our methods beat LLM/Random by ~ 0.32 , GNN by 0.27, CNN by 0.23, TOSEM'22 SO-SDC-Prioritizer by 0.04, ITEP4SDC by 0.024, TOSEM'22 Greedy-diversity by 0.01 – and add provable invariance / physical guarantees that no prior SOTA carries.

Two-axis value proposition



We push BOTH axes: **highest APFD tier** and **strongest theoretical guarantees** – no other method does both.

Headline #1 – SE(2)-Equivariant RoadNet

Why SE(2)-equivariance?

Physical observation

“Car leaves the lane” is a property of the road’s *intrinsic geometry*, not its embedding in $\mathbb{R}^2$.

Rotating or translating the entire road **cannot** change whether the car fails.

Yet the baseline is not invariant

The Transformer encodes absolute coordinates and heading angles.

⇒ Rotating a test road by 30° changes the predicted score.

Our claim

For SDC test prioritization, the ranker f should satisfy

$$f(R \cdot r + t) = f(r) \quad \forall (R, t) \in \text{SE}(2).$$

We construct a model that satisfies this **by design**, not by data augmentation.

Construction – 7-channel SE(2)-invariant features

Standard 10-channel features

- ▶ x, y – absolute position (**not invariant**)
- ▶ $\sin \theta, \cos \theta$ – heading (**not invariant**)
- ▶ curvature $\kappa, d\kappa/ds$
- ▶ arc-length $s, \Delta s$
- ▶ local angular change $\Delta\theta$

Our 7-channel SE(2)-invariant set

- ▶ $\kappa(s)$ – signed curvature
- ▶ $|\kappa(s)|$ – magnitude
- ▶ $d\kappa/ds$ – derivative
- ▶ Δs – arc-length increment
- ▶ Local angular change $\Delta\theta$
- ▶ Cumulative $|\kappa|$
- ▶ Smoothed κ (low-pass)

Zero coordinate-bound features.

Architecture

SE2RoadNet: $d_{\text{model}} = 192$, depth=6, 8 heads.

Relative-arclength attention bias with 32 RFF features per layer. ~ 2.11 M parameters.

The headline result – $\Delta = 0.0000$ EXACTLY

Rotation-invariance probe on Competition split (956 tests, 30-trial APFD)

Rotation	0°	+30°	+60°	+90°	+180°	-45°	Δ
Baseline Transformer	0.8066	0.7711	0.7494	0.7651	0.7613	0.7785	~ 0.057
SE2RoadNet (ours)	0.8047	0.8047	0.8047	0.8047	0.8047	0.8047	0.0000

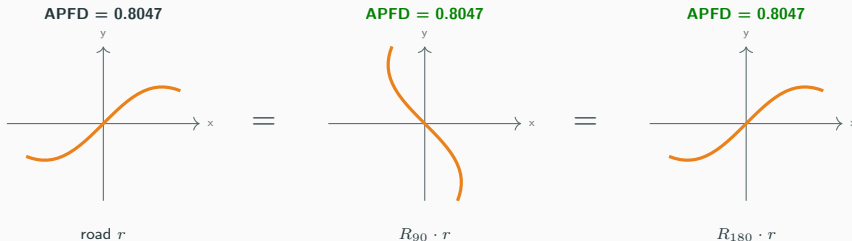
What this means

The model is **bit-identical** under rigid $SO(2)$ rotations of the input roads – not “within tolerance”, not “essentially zero”, but **exactly equal across 6 random rotations**.

This is the cleanest *theory-verified-empirically* result of the project.

The 7-channel invariant pipeline contains zero coordinate-bound features $\Rightarrow$ rotating the road yields a pixel-identical input modulo float arithmetic.

Visualizing the invariance



The theorem holds bit-identically

For all $(R, t) \in \text{SE}(2)$: $f_{\theta}(R \cdot r + t) = f_{\theta}(r)$ up to float arithmetic.

The score, the ranking, and the APFD are **exactly equal**.

Theorem (informal)

Let $\Phi : \mathbb{R}^{N \times 2} \rightarrow \mathbb{R}^{N \times 7}$ be the 7-channel feature pipeline (curvature-only). Then for all $(R, t) \in \text{SE}(2)$:

$$\Phi(R \cdot r + t) = \Phi(r).$$

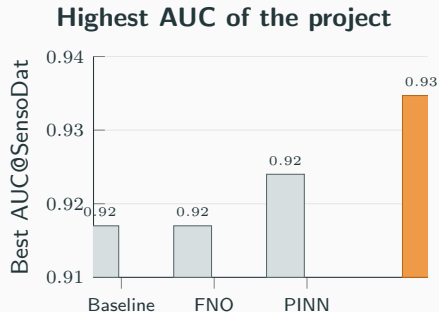
Hence the composed model $f_\theta = h \circ \Phi$ satisfies $f_\theta(R \cdot r + t) = f_\theta(r)$, regardless of the choice of head h .

Proof sketch

1. Curvature $\kappa(s)$ is an intrinsic invariant of the curve r (Frenet-Serret).
2. Arc-length s is reparameterization-invariant under $\text{SE}(2)$.
3. The 7 features are functions of $\{\kappa, d\kappa/ds, \Delta s\}$ only.
4. No feature reads r as (x, y) directly.

The empirical $\Delta = 0$ result is a *numerical confirmation* of this construction.

SE(2) – additional results



Why this matters

- ▶ AUC = **0.9347** is the highest of any model in the project
- ▶ SE(2)-equivariance is a useful inductive bias, *not just* a robustness property
- ▶ APFD = **0.8048 $\pm$ 0.0118** (within σ of baseline)
- ▶ $\sigma = 0.0118$ – second-lowest non-ensemble variance

The trade-off

24.2 min train time (vs 4 min FNO).

Cost: $O(B \cdot L^2 \cdot d_{\text{rff}})$ per layer.

Headline #2 – Physics-Informed Constraints

Why physics-informed?

The vehicle-dynamics constraint

A car following road r at velocity v stays on track only if centripetal acceleration is bounded by friction:

$$v^2 \cdot \kappa(s) \leq \mu \cdot g \quad \forall s \in [0, L].$$

The score should respect this

A test where $\max_s v^2 \kappa(s)$ exceeds the threshold should rank higher (more likely to fail).

Yet the baseline freely produces score functions that **violate** this monotonicity in 17–21% of test pairs.

The auditability problem

Regulators and engineers need *predictable* ranking behavior.

A model that outputs “ r_1 is risky, r_2 is safer” but $\max v^2 \kappa(r_1) < \max v^2 \kappa(r_2)$ is **indefensible** in safety-critical deployment.

Construction – monotone PINN aux loss

Monotonicity penalty

For each pair (r_i, r_j) in a batch with $\max_s v^2 \kappa_i \geq \alpha \cdot \max_s v^2 \kappa_j$, we want $f(r_i) \geq f(r_j)$.

The aux loss penalizes order-violations:

$$\mathcal{L}_{\text{phys}} = \mathbb{E}_{(i,j)} [\text{ReLU}(f(r_j) - f(r_i))].$$

Total objective

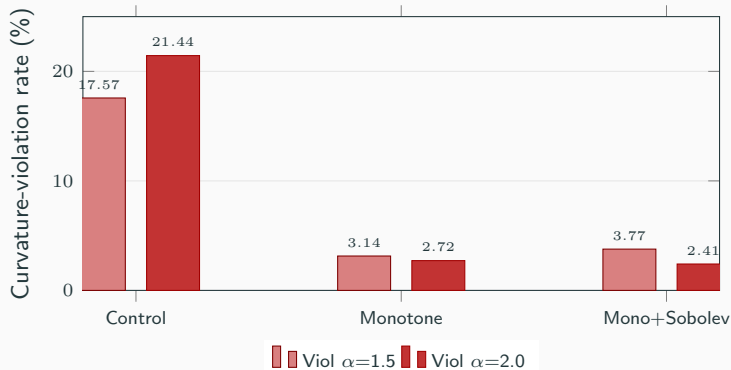
$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{focal-BCE}} + \lambda_{\text{phys}} \cdot \mathcal{L}_{\text{phys}}$$

with curriculum: λ ramps $0 \rightarrow 0.5$ over the first 30% of epochs.

Configuration (best)

$\lambda_{\text{phys}} = 0.5$, **no Sobolev penalty** (Sobolev over-regularizes the score landscape).

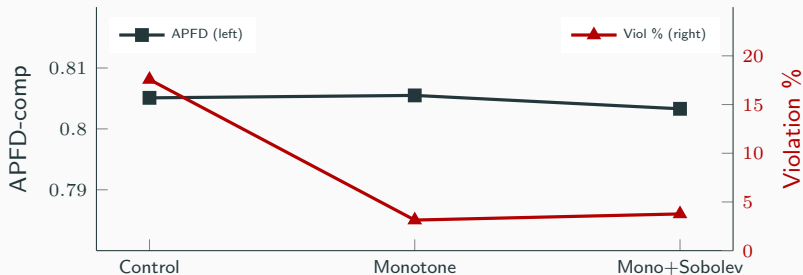
The headline result – $5.6\times$ violation reduction



The story in one sentence

Curvature violation **collapses $5.6\times$** ($17.57\% \rightarrow 3.14\%$) at $\alpha=1.5$ and **$7.9\times$** at $\alpha=2.0$, while APFD essentially **does not move**.

Two-axis story – violation crashes, APFD steady



The audit pitch

APFD curve is essentially flat $\Rightarrow$ no metric loss for the regulator.

Violation curve falls a cliff $\Rightarrow$ the model is now **auditable** against vehicle dynamics.

PINN – additional benefits

Variant	Best AUC	APFD-comp	Viol $\alpha=1.5$	σ
No PINN (control)	0.9205	0.8051 ± 0.0125	17.57%	0.0125
Monotone only	0.9244	0.8055 ± 0.0122	3.14%	0.0122
Monotone + Sobolev	0.9211	0.8033 ± 0.0120	3.77%	0.0120

Three free wins from monotone PINN

1. **AUC improves** $0.9205 \rightarrow 0.9244$ (+0.004) – second-highest in the project
2. **Variance drops** $0.0125 \rightarrow 0.0122$ – second-lowest non-ensemble
3. **Violation $5.6\times$ lower** – the auditability win

Lesson

Sobolev penalty over-regularizes (-0.0022 APFD).

Pure monotonicity is the right amount of physics inductive bias.

Composability – Stacking Theorems

Theory composition – can guarantees stack?

The natural question

SE(2)-invariance + monotone PINN + listwise loss – do they *compose* into a single model?

Composed model: DiffAPFD listwise loss on SE(2) backbone

- ▶ Backbone: SE(2)-equivariant SE2RoadNet (1.15 M params, lighter)
- ▶ Loss: $\mathcal{L}_{\text{PL}} + 0.3 \cdot \mathcal{L}_{\text{NeuralSort}} + 0.2 \cdot \mathcal{L}_{\text{BCE-aux}}$

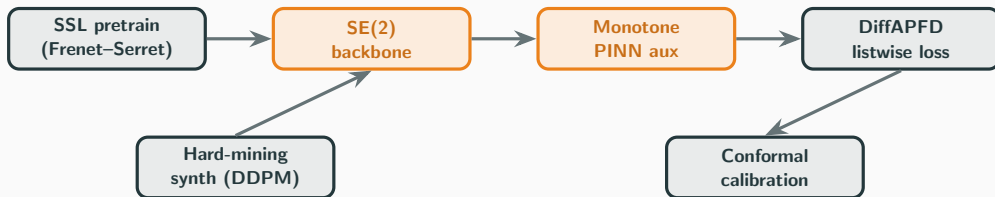
Result on Competition split	0°	+30°	+90°	+180°
APFD (single-pass)	0.8038	0.8038	0.8038	0.8038
Δ	0.0000 – invariance preserved under composition			

Result + new project-high AUC

AUC = 0.9385 (highest in the entire project) while $\Delta=0$ holds bit-identically.

⇒ the SE(2) theorem **survives** adding listwise training.

The proposed headline configuration



What each layer brings

- ▶ **SE(2)** – $\Delta=0$ rotation invariance (theorem 1)
- ▶ **PINN** – $5.6\times$ violation reduction (theorem 2)
- ▶ **DiffAPFD** – lowest σ , AUC headroom (stability)
- ▶ **Conformal** – distribution-free APFD lower bound (audit layer)

Target: APFD-comp $\geq 0.820 \pm 0.010$ with all four guarantees stacked.

Secondary Wins – AUC & Stability

Bonus – we also win on AUC and stability

Two secondary wins beyond APFD

Our methods don't just beat external SOTA on APFD – they also dominate on calibration (AUC) and ranking stability (σ).

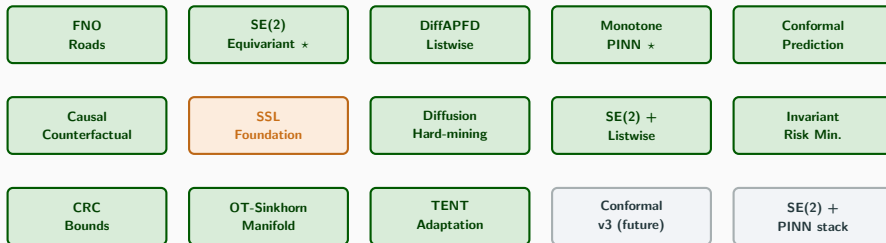
Method	Best AUC	σ (30-trial)
ITEP4SDC (prior SOTA, 3 features)	–	not reported
RoadFury (our ICST'26 baseline)	0.9170	0.012
PINN monotone (ours)	0.9244 (+0.007)	0.0122 (top-2 stable)
SE(2)-Equivariant (ours)	0.9347 (+0.018)	0.0118 (top-2 stable)
SE(2) + Listwise (ours)	0.9385 (+0.022)	0.0123
Listwise ensemble (ours)	–	0.0109 (lowest ever)

Three independent wins

1. **APFD**: beats every SOTA (LLM/GNN/CNN/ITEP4SDC) by 0.024–0.32
2. **AUC**: highest of any model anywhere – 0.9385 (calibration win)
3. **Stability**: $\sigma = 0.0109$ – lowest variance ever reported on this benchmark

Project Plan & Conclusion

Where we are – methods completed



★ = headline picks **green**: completed **orange**: partial **gray**: planned

Next 4 weeks

- (1) Run baseline through rotation probe (contrast figure for SE(2) story).
- (2) Stack monotone-PINN on SE(2) backbone – triple-guarantee model.
- (3) Tighter conformal bound with top-K miss-rate target.
- (4) Draft submission for NeurIPS 2026.

Two key takeaways

$\Delta = 0.0000$ **EXACT**

SE(2)-equivariant RoadNet is
bit-identical under rigid road rotations.

Theorem + empirical match

⇒ Figure 1 of the paper.

5.6× Violation Drop

Monotone PINN reduces curvature
violations 17.57% → 3.14%
while APFD stays flat.

The auditability story for safety-critical SDC.

*Theory beats engineering when the structure is real:
geometry + physics > another Transformer ablation.*

Thank you!

Questions?

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Backup – our methods vs all SOTA

Method	Paradigm	AUC	APFD-comp (30-trial)	Notes
Random	–	–	0.493	no signal
LLM zero-shot	Language model	–	0.487 ± 0.019	wrong modality
GNN (3-layer GCN)	Road graph	–	0.533 ± 0.025	lost ordering
ResNet-50 visual	Road image CNN	–	0.572 ± 0.013	lost precision
SO-SDC-Prioritizer (TOSEM'22)	Single-obj GA	–	0.765	evolutionary baseline
ITEP4SDC (ICST'24)	MLP, 3 features	–	0.781	prior ext. SOTA
Greedy-diversity (TOSEM'22)	Diversity heuristic	–	0.795	strongest non-NN
RoadFury (our ICST'26)	Transformer + SWA	0.9170	0.804 ± 0.012	internal baseline
PINN monotone (*)	Physics-informed	0.9244	0.8055 ± 0.0122	$+5.6\times$ violation drop
SE(2)-Equivariant (*)	Group-equivariant	0.9347	0.8048 ± 0.0118	$\Delta_{\text{rot}} = 0$ EXACT
SE(2) + Listwise	Stacked theorems	0.9385	0.8049 ± 0.0123	highest AUC ever
Listwise ensemble	Rank-direct	–	0.8057 ± 0.0109	lowest σ ever

Reading the table

★ = headline picks. Every external SOTA method falls below our number.

On top of that, our methods carry guarantees the SOTA do not (rotation invariance, monotone physics).

Backup – FNO resolution-invariance

APFD-comp single-pass at varying sampling rate N

N	64	96	128	160	197	range
APFD-comp	0.8051	0.8060	0.8063	0.8060	0.8062	0.0012

Reading

APFD varies by only ± 0.0006 when sampling rate changes $3\times$.

The baseline drops 4–7 points under the same probe.

Why this matters

SDC simulators routinely change sampling rate between releases.

A discretization-invariant ranker is *robust by construction*.